Deep Learning in Survival Analysis: Recent Advances and Multimodal Approaches

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Abstract

Survival analysis is the statistical methodology for time-to-event data in the presence of censoring, where event times may be only partially observed. This review provides a structured overview of recent developments in deep learning–based survival modeling, with emphasis on statistical foundations and modeling assumptions. The discussion begins with the core functionals of survival analysis, including the survival function, hazard function, and cumulative hazard, together with the independent censoring assumption required for identifiability in classical semi-parametric models such as the Cox proportional hazards model. Subsequent sections describe how neural network–based approaches generalize these frameworks by parameterizing flexible hazard or survival functions while retaining censoring-aware likelihood principles. Commonly used objective functions—including partial likelihood, discrete-time likelihood formulations, and inverse probability–weighted losses—are interpreted in relation to their underlying probabilistic assumptions. The review further summarizes multivariate and multi-task survival models, copula-based dependence structures, and generative latent-variable approaches that yield full predictive distributions. Finally, recent extensions to causal survival analysis and dynamic treatment regime learning are discussed, including counterfactual Q-learning and Buckley–James–based methods for censored outcomes. Open issues include identifiability under complex censoring mechanisms, principled uncertainty quantification (including bootstrap methods), and the integration of large-scale foundation models within survival frameworks.

Keywords: Survival Analysis; Censoring; Cox Proportional Hazards Model; Deep Learning; Causal Survival Analysis.

1 Introduction

Survival analysis is the statistical methodology for time-to-event data in the presence of right censoring. Core concepts, including censoring mechanisms and the Cox proportional hazards model, are treated in the corresponding encyclopedia entries [1, 2]. This article focuses on modern extensions based on deep learning and related computational approaches.

In survival data, the event time may not be fully observed. Each individual contributes a follow-up time indicating the duration of observation and an indicator specifying whether the event occurred. When the event is not observed within the study period, the observation is right-censored. The resulting data structure consists of follow-up times, event indicators, and covariates.

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Statistical inference requires assumptions on the censoring mechanism. The standard assumption is independent (non-informative) censoring, meaning that conditional on covariates, the event time and censoring time are independent [3, 4]. Under this assumption, the event-time distribution is identifiable.

Event-time distributions can be characterized equivalently by the survival function, the hazard function, or the cumulative hazard function [5]. These representations are mathematically equivalent, and modeling strategies correspond to different parameterizations. Further details are provided in the relevant encyclopedia entries.

The Cox proportional hazards model [6] is the standard semi-parametric regression model for censored data. It assumes that covariates act multiplicatively on an unspecified baseline hazard function. The defining structural assumption is proportional hazards, implying constant hazard ratios over time. Estimation is based on the partial likelihood, which depends only on event ordering and yields consistent estimators under independent censoring [3]. A dedicated entry provides a detailed treatment of this model.

Although widely used, the proportional hazards assumption may be restrictive in high-dimensional or nonlinear settings [7, 8]. This limitation has motivated flexible machine-learning extensions.

Deep survival models generalize classical approaches by replacing linear predictors with nonlinear function approximators such as neural networks [9]. Many formulations are derived from likelihood principles, including extensions of the partial likelihood and discrete-time likelihood approaches [10, 11]. Censored observations contribute information through survival up to the censoring time.

These methods extend naturally to multivariate survival analysis and dependence modeling [12, 13]. Copula-based formulations allow separate specification of marginal event-time distributions and their dependence structure, as described in the encyclopedia entry on copula models [14].

Right censoring may also be interpreted within missing data theory. Observed events correspond to fully labeled outcomes, whereas censored observations provide partial information. This perspective connects survival analysis to positive–unlabeled learning [15, 16]. When censoring depends on unobserved variables, additional modeling assumptions are required, relating the problem to missing-not-at-random frameworks in missing data theory [17].

Survival methods also play an important role in causal inference. In this setting, the objective is to estimate treatment effects on time-to-event outcomes under censoring [18]. Recent approaches integrate survival models with dynamic treatment regimes and reinforcement learning frameworks. Counterfactual Q-learning and Buckley–James–based procedures extend regression methods to sequential decision problems with censored outcomes [19, 20]. These developments connect survival analysis with potential outcomes and policy learning.

Overall, modern survival analysis includes semi-parametric modeling, deep learning extensions, multivariate dependence structures, and causal decision frameworks. The central statistical problem remains the estimation of event-time distributions from partially observed data under censoring, with explicit attention to identifiability assumptions and consistent inference.

2 Mathematical Framework and Notation

Deep survival modeling concerns estimation of event-time distributions from right-censored data. In this setting, the outcome of interest is not fully observed for all individuals.

2.1 Observation Structure and Identifiability

For each individual $i = 1, \ldots, n$, let T_i^* denote the true event time and C_i the censoring time. The observed data consist of

$$T_i = \min(T_i^*, C_i), \qquad \delta_i = \mathbb{I}(T_i^* \leq C_i),$$

together with a covariate vector $X_i \in \mathbb{R}^p$.

Here, T_i is the recorded follow-up time and δ_i indicates whether the event occurred ($\delta_i = 1$) or the observation is right-censored ($\delta_i = 0$). The observed sample is

$$\mathcal{D} = \{(T_i, \delta_i, X_i)\}_{i=1}^n,$$

typically assumed to be independent and identically distributed.

Statistical inference relies on assumptions about the censoring mechanism. The standard condition is independent (non-informative) censoring,

$$T_i^* \perp C_i \mid X_i,$$

which states that, conditional on covariates, the event time and censoring time are independent. Under this assumption, the distribution of T^* is identifiable from the observed data [3].

2.2 Event-Time Distribution

Let T^* denote the event-time random variable and let X denote covariates. The conditional distribution of T^* given $X = x$ can be characterized by the following equivalent functionals [5].

The survival function is

$$S(t \mid x) = \mathbb{P}(T^* > t \mid X = x).$$

The hazard function is

$$\lambda(t \mid x) = \lim_{\Delta t \to 0} \frac{\mathbb{P}(t \leq T^* < t + \Delta t \mid T^* \geq t, X = x)}{\Delta t}.$$

The cumulative hazard function is

$$\Lambda(t \mid x) = \int_0^t \lambda(u \mid x)\, du.$$

These quantities satisfy the relationships

$$S(t \mid x) = \exp\{-\Lambda(t \mid x)\}, \qquad f(t \mid x) = \lambda(t \mid x) S(t \mid x),$$

where $f(t \mid x)$ denotes the event-time density when it exists.

Thus, specifying any one of these functions determines the others.

2.3 Likelihood Under Censoring

Under independent censoring, the likelihood contribution of individual i can be written as

$$L_i = \left[f(T_i \mid X_i)\right]^{\delta_i} \left[S(T_i \mid X_i)\right]^{1-\delta_i}.$$

If $\delta_i = 1$, the observation contributes the event-time density; if $\delta_i = 0$, it contributes the survival probability evaluated at the censoring time.

This likelihood structure forms the basis of classical semi-parametric methods as well as modern deep learning–based survival models.

2.4 Fundamental Representations of the Event-Time Distribution

Let T^* denote the event-time random variable and X the covariate vector. The conditional distribution of T^* given $X = x$ may be characterized by three equivalent functionals [5].

The *survival function* is defined as

$$S(t \mid x) = \mathbb{P}(T^* > t \mid X = x),$$

representing the probability that the event has not occurred by time t.

The *hazard function* is defined as

$$\lambda(t \mid x) = \lim_{\Delta t \to 0} \frac{\mathbb{P}(t \le T^* < t + \Delta t \mid T^* \ge t, X = x)}{\Delta t},$$

which describes the instantaneous event rate at time t conditional on survival up to that time.

The *cumulative hazard function* is

$$\Lambda(t \mid x) = \int_0^t \lambda(u \mid x)\, du.$$

These quantities satisfy the relationships

$$S(t \mid x) = \exp\{-\Lambda(t \mid x)\}, \qquad \lambda(t \mid x) = -\frac{\partial}{\partial t} \log S(t \mid x).$$

Hence, specifying any one of these functions determines the others. Different survival models correspond to alternative parameterizations of these equivalent representations.

2.5 Neural Extensions of Proportional Hazards Models

The Cox proportional hazards model [6] is the standard semi-parametric regression framework for right-censored data. It specifies the conditional hazard function as

$$\lambda(t \mid X) = \lambda_0(t) \exp(\beta^\top X),$$

where $\lambda_0(t)$ is an unspecified baseline hazard function and β is a vector of regression coefficients.

The defining feature of the model is the proportional hazards property:

$$\frac{\lambda(t \mid X_1)}{\lambda(t \mid X_2)} = \exp\{\beta^\top (X_1 - X_2)\},$$

which is independent of time t. Consequently, covariate effects act multiplicatively on the baseline hazard and induce time-constant hazard ratios.

Estimation of β is based on the partial likelihood, which depends on the ordering of event times and does not require explicit specification of $\lambda_0(t)$. Under independent censoring, the resulting estimators are consistent and asymptotically normal [3].

Neural proportional hazards models, such as DeepSurv [9], extend this framework by replacing the linear predictor with a nonlinear function $h_\theta(X)$:

$$\lambda(t \mid X) = \lambda_0(t) \exp\{h_\theta(X)\}.$$

This formulation preserves the proportional hazards structure while allowing flexible covariate effects. Such models are primarily used for risk ranking and prognostic scoring.

2.6 Discrete-Time and Non-Proportional Models

Discrete-time survival models, including DeepHit [10], divide the time axis into ordered intervals

$$0 < t_1 < t_2 < \cdots < t_K.$$

Event-time modeling is then carried out on this discrete grid.

Let

$$p_k(x) = \mathbb{P}(T^* = t_k \mid X = x),$$

denote the conditional probability that the event occurs in interval t_k. The vector

$$\{p_1(x), \ldots, p_K(x)\}$$

constitutes a conditional probability mass function over time. Survival probabilities can be obtained by cumulative summation of these probabilities.

Because the model specifies time-specific probabilities directly, covariate effects may vary across intervals. This formulation therefore accommodates non-proportional hazards and permits flexible temporal dynamics.

These discrete-time approaches extend naturally to competing risks settings by defining separate probability distributions for each event type, while ensuring that the probabilities across all risks and time intervals sum to one.

2.7 Causal Survival and Dynamic Treatment Regimes

In clinical applications, survival analysis may be integrated with causal inference to evaluate treatment strategies under right censoring. The inferential target is no longer solely the event-time distribution, but the effect of sequential interventions on long-term outcomes.

Let A_t denote a treatment assigned at time t, and let S_t represent the corresponding state, summarizing current covariates and historical information. Under a policy π, the sequence of actions is determined by $\pi(a \mid s)$. The associated action-value function is defined as

$$Q^\pi(s, a) = \mathbb{E}^\pi \left[\sum_{k=t}^{\infty} \gamma^{k-t} R_k \;\middle|\; S_t = s, A_t = a \right],$$

where R_k denotes the reward at time k and $\gamma \in (0, 1)$ is a discount factor. The function $Q^\pi(s, a)$ represents the expected discounted cumulative reward obtained by taking action a in state s and subsequently following policy π.

In survival settings, rewards are typically defined in relation to event times, for example through survival indicators, cumulative hazard reductions, or functions of time-to-event outcomes. Because censoring may truncate observed trajectories, estimation procedures incorporate appropriate weighting or imputation adjustments.

Counterfactual Q-learning and Buckley–James–based extensions adapt classical regression and dynamic programming ideas to censored sequential data [19, 20]. These methods combine survival modeling, potential outcomes theory, and policy optimization to estimate dynamic treatment regimes that target long-term survival objectives.

2.8 Generative and Multimodal Approaches

Generative survival models represent event times using latent variables to account for unobserved heterogeneity. Let Z denote a latent variable capturing unmeasured risk factors or subgroup structure. The joint distribution of the event time and covariates can be expressed through a latent-variable decomposition [21],

$$p(T^*, X) = \int p(T^* \mid Z, X)\, p(Z \mid X)\, dZ.$$

This formulation separates observed predictors from latent structure and yields a full probabilistic model for event times. Because the complete distribution is specified, such models naturally provide predictive survival curves together with uncertainty quantification.

In multimodal settings, data may include images, molecular measurements, and longitudinal clinical records. Transformer-based architectures [22] are commonly used to integrate these heterogeneous sources through attention mechanisms. Given query, key, and value matrices Q, K, and V, attention is computed as

$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V,$$

where d denotes the feature dimension. The attention weights determine how information from different components is combined into a unified representation.

These mechanisms support flexible fusion of imaging, genomics, and electronic health record data within survival prediction frameworks, while preserving a coherent probabilistic or representation-learning structure.

2.9 Conceptual Synthesis

Despite differences in modeling architecture and estimation strategy, modern survival approaches share a common statistical objective: the recovery of event-time distributions from partially observed data subject to right censoring. Whether formulated as semi-parametric regression, discrete-time probability modeling, generative latent-variable modeling, or causal policy learning, each method addresses the same underlying inferential problem of incomplete outcomes.

Classical likelihood principles continue to provide the foundational framework for estimation and identifiability. Deep learning extensions primarily act as flexible parameterizations of survival functionals, allowing complex nonlinear relationships, high-dimensional covariates, and multimodal data sources to be incorporated within censoring-aware statistical models. In this sense, modern methods extend rather than replace the core probabilistic structure of survival analysis.

3 Materials and Methods

This section describes the computational framework for multimodal survival analysis and the associated statistical objectives. The overall approach follows censoring-aware learning principles grounded in classical survival theory [3, 5].

3.1 Data Modalities and Representation Learning

Modern survival applications frequently combine heterogeneous data sources, including imaging, molecular profiles, and longitudinal clinical records. Each modality is transformed into a numerical representation prior to integration.

3.1.1 Whole Slide Images

Whole slide images (WSIs) are high-resolution digital pathology images that are computationally infeasible to process directly. Standard practice divides each slide into smaller patches and applies deep feature extraction to each patch. Vision Transformer (ViT) architectures provide scalable self-attention mechanisms for encoding spatial structure [22].

Attention-based aggregation of patch-level features corresponds to Multiple Instance Learning (MIL), where patient-level supervision is available but patch-level labels are not [23]. The resulting weighted representation yields a patient-level embedding that summarizes prognostically relevant regions.

3.1.2 Omics Data

High-dimensional molecular measurements, such as gene expression profiles, require normalization and dimensionality reduction prior to modeling. Standard preprocessing includes scale normalization to ensure comparability across samples.

Because the number of features often exceeds the number of individuals, latent-variable methods such as autoencoders and Variational Autoencoders (VAEs) are used for representation learning [21]. These architectures learn low-dimensional embeddings that capture dominant biological variation while reducing noise and collinearity.

3.1.3 Electronic Health Records

Electronic health records (EHRs) consist of longitudinal and irregularly sampled clinical observations. Recurrent neural networks, including Gated Recurrent Units (GRUs), are commonly used to model temporal dependence in such data [13]. These models generate time-aware patient embeddings that summarize historical trajectories.

3.2 Multimodal Integration

Modality-specific embeddings are integrated into a unified representation using either late fusion (e.g., concatenation) or shared latent spaces that allow cross-modal interaction. Transformer-based attention mechanisms provide a flexible framework for multimodal alignment [24].

The resulting representation serves as input to survival models.

3.3 Survival Modeling Objectives

Integrated representations are trained using censoring-aware objectives consistent with classical likelihood theory [3]. Common approaches include:

partial likelihood methods derived from the Cox proportional hazards model [6];

discrete-time likelihood formulations for interval-based event modeling [10];

fully probabilistic continuous-time models that directly parameterize the survival function.

Each objective corresponds to a distinct representation of the event-time distribution and relies on appropriate assumptions regarding the censoring mechanism.

3.4 Evaluation Protocol and Metrics

Model evaluation follows standard procedures for censored data [25]. Performance is assessed using:

the concordance index (C-index), which measures risk ranking accuracy;

the Brier score, which evaluates probabilistic calibration;

time-dependent area under the ROC curve, which quantifies discrimination at specific horizons [26].

These metrics account for censoring and are widely used in survival model comparison.

3.5 Subgroup Fairness

Performance disparities across demographic or clinically relevant subgroups are evaluated using subgroup-specific metrics. Disparity may be summarized by the range of subgroup performance measures. Such analyses align with emerging literature on fairness in predictive modeling [27].

3.6 Causal Survival Q-Learning Extension

In causal settings, the objective extends from prediction to policy optimization under censoring. Dynamic treatment regimes formalize sequential decision-making in time-to-event contexts [28].

Inverse probability of censoring weighting (IPCW) adjusts for incomplete follow-up by reweighting observations according to the estimated probability of remaining uncensored [29]. This approach enables consistent estimation under independent censoring and forms the basis for reinforcement learning extensions such as counterfactual Q-learning in survival settings [19].

3.7 Comparative Perspective

Deep survival methodologies may be organized into three broad paradigms:

(1) Semi-parametric ranking-based models derived from the Cox framework; (2) Discrete-time probability models; (3) Causal and reinforcement learning–based prescriptive approaches.

Although these paradigms differ in inferential targets, structural assumptions, and computational strategies, they share a common objective: estimation of time-to-event quantities from censored data under clearly specified identifiability conditions. Their differences are summarized in Table 1.

Cox-based neural models such as DeepSurv extend the classical proportional hazards framework by replacing the linear predictor with a nonlinear function approximator. The primary inferential target remains relative risk ranking, typically evaluated through concordance-type measures. Because the baseline hazard function is not explicitly modeled, estimation retains a semi-parametric structure and remains computationally efficient. These models inherit the proportional hazards assumption, implying constant hazard ratios over time. When this assumption is violated—such as in the presence of time-varying effects or crossing survival curves—model adequacy may be affected.

Discrete-time survival models such as DeepHit relax proportionality by modeling the conditional event probability over a partitioned time axis. This formulation estimates the full event-time distribution within each interval and naturally accommodates non-proportional hazards and competing risks. A central methodological consideration is the time discretization scheme. Coarse grids reduce temporal resolution, whereas overly fine grids may result in sparse event counts and increased estimation variability. Thus, model performance depends on appropriate alignment between temporal resolution and data density.

Causal and reinforcement learning–based approaches represent a further extension of survival methodology. Rather than focusing exclusively on prediction, these frameworks aim to learn decision rules that optimize long-term outcomes under time-to-event data. They incorporate potential outcomes theory, sequential treatment assignment, and counterfactual reasoning. Additional components typically include confounding adjustment and censoring correction. While these approaches broaden the scope of survival analysis toward policy learning, they require stronger identification assumptions and involve more complex estimation procedures.

Table 1: Comparative Summary of Major Deep Survival Paradigms

Feature	**Cox-Type Neural Models**	**Discrete-Time Survival Models**	**Causal / Policy-Based Models**
Primary Objective	Relative risk estimation and ranking	Direct estimation of event probabilities over time	Optimization of treatment policies or decision rules
Time Structure	Continuous time	Discretized time intervals	Continuous or discrete, policy-dependent
Hazard Assumption	Proportional hazards structure	Time-varying effects permitted	Not restricted to proportionality; depends on model specification
Censoring Treatment	Partial likelihood under independent censoring	Likelihood-based modeling of time intervals	Weighting or imputation methods (e.g., IPCW, model-based adjustments)
Dependence / Multivariate Extension	Requires additional modeling	Naturally extended to competing risks	Counterfactual multi-event or sequential formulations
Inferential Target	Hazard ratios	Survival probabilities	Value functions or expected long-term outcomes
Computational Structure	Single forward pass with risk score output	Vector-valued probability estimation	Iterative optimization (e.g., Bellman-type updates)
Typical Application Context	Prognostic risk stratification	Time-specific prediction and competing risks analysis	Dynamic treatment regimes and sequential decision-making

Collectively, these paradigms illustrate a conceptual progression from relative risk modeling, to full distribution estimation, and ultimately to intervention-oriented survival learning.

Table 2 summarizes major methodological developments in deep survival analysis from 2018 to 2026, highlighting the progression from predictive modeling to multimodal representation learning and causal policy frameworks.

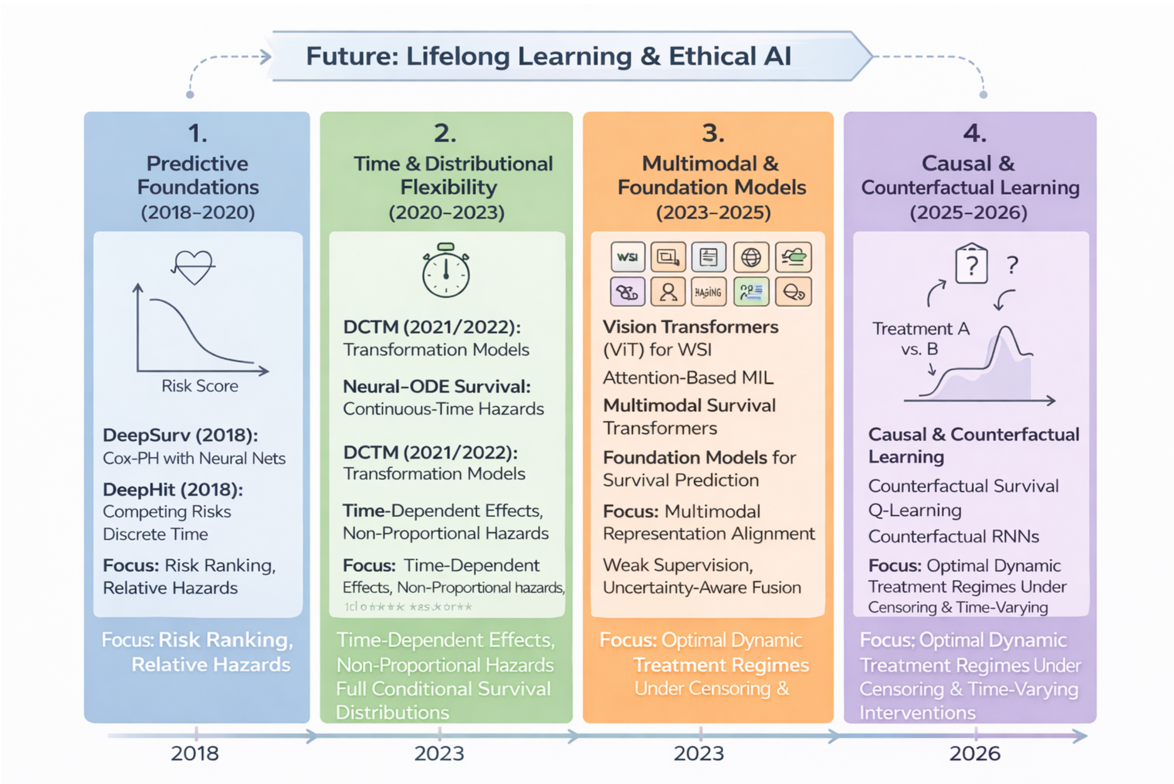

Figure 1: Evolutionary development of deep survival analysis (2018–2026). The field may be viewed as progressing through four conceptual phases: (1) Predictive Foundations, characterized by neural extensions of Cox proportional hazards models for risk ranking and relative risk estimation; (2) Distributional Flexibility, introducing models that relax proportional hazards assumptions and estimate full conditional survival distributions using transformation-based and continuous-time architectures; (3) Multimodal and Foundation Models, integrating electronic health records, medical imaging, pathology whole-slide images, and omics data through attention mechanisms and Transformer-based representation learning; and (4) Causal and Counterfactual Frameworks, emphasizing heterogeneous treatment effect estimation, counterfactual reasoning, and optimization of dynamic treatment regimes under censoring.

Table 2: Major methodological developments in deep survival analysis

Era & Theme	Representative Milestones	Primary Focus & Methodological Advances
2018–2020 Predictive Foundations	DeepSurv [9] DeepHit [10] Neural Cox models [11]	Neural extensions of Cox proportional hazards; discrete-time survival modeling with competing risks; emphasis on concordance and relative risk prediction.
2020–2023 Distributional Flexibility	DCTM [30] Neural-ODE Survival Time-dependent ROC [26]	Modeling non-proportional and time-varying effects; learning full conditional survival distributions via transformation models and continuous-time dynamics.
2023–2025 Multimodal & Foundation Models	Vision Transformers [22, 24] Multimodal survival transformers	Integration of EHR, imaging, pathology WSI, genomics, and omics data using Transformers and attention-based MIL; weak supervision and representation alignment.
2024–2025 Copula & Multi-task Learning	Copula DL Survival [12] Multi-task CNN–LSTM [13]	Joint modeling of correlated time-to-event and count outcomes; copula-based dependence learning; structured multitask survival modeling under censoring.
2025–2026 Causal & Policy Learning	Counterfactual Survival Q-Learning [19, 20] Causal Survival DL for competing risks [31]	Estimation of heterogeneous treatment effects under censoring; counterfactual reasoning; optimization of dynamic treatment regimes.

4 Discussion and Future Directions

The development of modern survival analysis reflects increasing interaction between classical time-to-event methodology, high-dimensional representation learning, and causal inference. Traditional models, including the Cox proportional hazards model, remain foundational for risk estimation and regression with censored data. Recent research extends these frameworks toward more flexible distributional modeling and decision-oriented applications.

Foundation Models Large-scale pretrained architectures, commonly referred to as foundation models, are increasingly applied in biomedical settings, including genomics and medical imaging. Relevant background is provided in the encyclopedia entry on foundation models in healthcare. Within survival analysis, such models may serve as pretrained representation learners that are adapted to time-to-event objectives through fine-tuning with censoring-aware loss functions. Statistically, this approach can be interpreted as a form of structured regularization that leverages external data to improve stability and generalization.

Causal Survival Inference Survival analysis is closely connected to causal inference when the objective is to estimate treatment effects on time-to-event outcomes. This includes estimation of counterfactual survival under alternative interventions and the learning of optimal dynamic treatment regimes. These topics are discussed in the relevant entries on causal inference and sequential decision-making. Methods such as doubly robust estimation provide protection against certain forms of model misspecification by combining outcome modeling with weighting adjustments. In survival settings, these techniques are adapted to accommodate right censoring.

Uncertainty Quantification Reliable uncertainty quantification is essential when survival predictions inform clinical decisions. Standard approaches include bootstrap resampling, as described in the encyclopedia entry on bootstrap methods, as well as Bayesian and ensemble-based modeling strategies. These methods provide interval estimates or posterior distributions for survival quantities, enabling assessment of estimation variability and model stability.

Fairness and Robustness In applied settings, survival models should be evaluated for performance consistency across relevant subgroups, such as demographic or clinical strata. Disparity analysis examines whether predictive accuracy varies across groups. Robust optimization and distributionally robust approaches have been proposed to improve stability under population heterogeneity and potential distributional shift. These considerations are increasingly discussed within the broader literature on fairness in statistical learning.

Limitations Despite methodological progress, several limitations remain. Deep survival models typically require substantial data for reliable estimation and may exhibit sensitivity to model specification and hyperparameter choices. Interpretability may be reduced relative to classical semi-parametric models. In causal survival analysis, identification relies on assumptions such as exchangeability and positivity, which are not directly testable from observed data. These constraints are inherent to observational inference and are discussed in the corresponding encyclopedia entries.

Concluding Perspective Contemporary survival analysis integrates classical semi-parametric theory with modern computational methods, including deep learning and causal decision frameworks. Across these developments, the central statistical objective remains the estimation of event-time distributions under censoring, together with transparent specification of assumptions and principled inference. Modern extensions expand the scope of survival methodology while preserving its foundational emphasis on identifiability and consistency.

Entry Navigation For risk ranking methods, see the entries on neural proportional hazards models. For non-proportional and discrete-time formulations, see entries on time-varying survival models. For treatment selection and dynamic decision frameworks, see entries on causal inference and reinforcement learning.

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